

Executive Summary

Open Neighbourhoods Precede Concept Spread: Structural Antecedents and Cross-Disciplinary Diffusion of Emerging Scientific Concepts

September 2026

Research Questions and Data

This study investigates how emerging scientific concepts spread across disciplinary boundaries. Two research questions structure the work:

RQ1 Which temporal and structural patterns in an evolving scientific knowledge graph characterise concept emergence?

RQ2 How do emerging concepts diffuse across disciplinary communities, and which network patterns distinguish locally concentrated from broadly integrated concepts?

The study uses 426 semantically grounded concepts and 462,812 works from OpenAlex. The main pool of 366 concepts is split into a 247-concept screen fold and a 119-concept held-out fold; a separate 191-concept MeSH population provides independent replication. Twenty-five yearly co-occurrence network snapshots ($\sim 27,000$ nodes, $\sim 84,000$ edges) are constructed with Leiden community detection.

Key Findings

1. Host-Entry Grafting Predicts Durable Integration (RQ2 — Confirmed)

When a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors. The effect replicates across all folds:

Fold	IRR/SD	95% CI	<i>p</i>
Screen (co-primary)	1.30	[1.16, 1.45]	10^{-5}
Held-out (co-primary)	1.19	[1.06, 1.33]	0.004
MeSH (co-primary)	1.23	[1.12, 1.36]	< 0.001
IVW pooled	1.26	[1.17, 1.36]	—

Co-transfer (arriving with origin companions) is null across all specifications.

The effect operates through **both margins**: host-leaning entry vocabulary raises the probability that uptake starts (extensive, IVW +6.43 pp/SD) and its magnitude given it has started

(intensive, IVW IRR/SD 1.183). It is **host-specific** (generality controls leave the coefficient unchanged; IVW retention 0.969) and shows **no field boundary** (Wald $p = 0.638$). The mechanism is host-vocabulary: topic-score controls absorb only -7.3% of the log-IRR (CI includes zero).

2. Persistent-Neighbour Closure Before Emergence (RQ1 — Partially Confirmed)

The primary closure measure did not survive held-out confirmation (R1_DEAD: Holm $p = 0.147$). However, **persistent-neighbour closure** — triadic closure among stable top-20 associates — is confirmed: held-out $S = -1.07$ $[-1.86, -0.29]$, Holm $p = 0.015$. The effect is not Burt brokerage (constraint is higher, not lower, on both folds).

3. Diffusion Typology

Only $k = 2$ is stable (min Jaccard 0.861): **localised** ($n = 136$) concepts stay near-monopolised by a single subfield, and **broad-from-the-start** ($n = 66$) concepts split across subfields from their first window. Finer typologies are unstable. In 25 of 26 pooled main-arm dual-onset concepts, network expansion precedes disciplinary diffusion.

Dead Ends

- **Source-sink viability.** Blocked by Gate A failure: only 17.3% of host edges meet the within-host citation tracing threshold (vs 64.2% under lenient any-parent tracing).
- **Breadth prediction.** Openness does not predict where concepts travel beyond growth baselines (all Holm $p = 0.405$).
- **Closure-anchoring link.** No detectable association between pre-emergence closure and later host-entry anchoring (partial $\rho = 0.049$, $p = 0.56$).
- **Co-transfer.** Null across all folds and populations.
- **Community roles.** BRIDGE role does not predict host entry (OR 0.64 [0.38, 1.08]).

Caveats and Open Questions

1. The concept pool is dominated by physics, physical sciences, and computer science; social sciences and humanities are untested.
2. The primary specification (concept $\times$ entry-year + host $\times$ entry-year FE) remains inconclusive on the held-out fold (IRR/SD 0.98, $G = 30$ clusters).
3. 87% of co-primary events are single-paper entries; on the held-out fold, the single-paper interaction ratio is 0.80 ($p = 0.004$).
4. The binary establishment threshold (EST_bin) is null on held-out ($p = 0.165$).
5. The study design is observational; host-nativeness of entry partners is not randomly assigned.
6. CRV1 rejection rate under the null is anti-conservative (17.5%); wild-cluster bootstrap $p = 0.012$ on the held-out fold.

Adopter-Level Mechanism

Adopters are $3.1\times$ more likely than matched non-adopters to have prior exposure to the concept's entry partners (OR 3.09 [2.32, 4.28]). The gradient runs FOREIGN > ADJACENT > NATIVE (OR 2.88 > 1.91 > 1.47). This runs opposite to the entry-level finding: at the entry level, host-native partners lower the barrier for newcomers lacking origin expertise; at the adopter level, those who adopt tend to have origin expertise already.

Run Summary

Five iterations produced 25 artifacts (5 datasets, 9 experiments, 9 evaluations, 2 research dossiers). A full numerical audit (Iteration 5) verified 99.4% curated-number accuracy (319 numbers) and closed all 6 major reviewer critiques. The target venue is Applied Network Science.